

Syllabus

Department	Math & CS	Credits	3
Class Room	Changui-gwan 5701	Subject	MC0000: Machine Learning & Deep Learning
Instructor	Sumin Han	Class	1
Lab (e-mail)	suminhan@ksa.hs.kr	hrs/wk	3

1. Course Description

Course Objectives

On completion of this course, you'll hopefully be able to:

- Understand and implement classical supervised learning (regression, classification) and unsupervised learning (clustering, dimensionality reduction) methods.
- Correctly apply model selection and validation principles (regularization, cross-validation) to real data.
- Understand the training principle behind neural networks (backpropagation) and design/implement representative deep learning architectures such as CNNs, RNNs, and Transformers.
- Understand the basic principles of generative models and large language models, and follow the current landscape of AI technology.
- Carry out the full pipeline of a machine learning project — problem definition, implementation, honest validation, and presentation — as a team.

Contents

- Regression models (linear regression, logistic regression)
- Generative-model view of classification (Naive Bayes, GDA)
- Distance-based models and clustering (kNN, k-means, GMM/EM)
- Support Vector Machines and kernels
- Regularization and model selection (Lasso/Ridge, cross-validation)
- Tree-based models (decision trees through GBDT)
- **[Block A Capstone] Team project: classical machine learning**
- Neural network basics, backpropagation, training techniques
- Convolutional Neural Networks (CNNs)
- Sequence models (RNNs)
- Attention and Transformers
- Graph representation learning (GNNs)
- Representation learning (PCA, word2vec, embeddings)
- Latent-variable generative models (VAE, GAN, Diffusion)
- **[Block B Capstone] Team project: deep learning**

2. Text & References

Course materials: instructor-prepared lecture notes and slides.

- “Machine Learning in Action”, Peter Harrington
- “Machine Learning Design Patterns”, V. Lakshmanan, S. Robinson, M. Munn
- “Deep Learning”, Ian Goodfellow, Yoshua Bengio, Aaron Courville

3. Grading

Activities	Quizzes	Homework	Midterm	Final	Team Project	Attendance
Percentages	10%	10%	25%	25%	25%	5%

- **Quizzes:** a short True/False quiz each week checking conceptual understanding of that week's material.
- **Homework:** individual programming assignments implementing the algorithms covered in class.
- **Midterm / Final:** written exams covering the theory of Block A and Block B respectively.
- **Team Project:** two capstone projects (Week 8: classical ML, Week 16: deep learning), each graded on a written report, a presentation, and peer-review. The grading emphasizes the *honesty of the experimental procedure* (baseline comparisons, proper train/validation/test separation, honest reporting of results) over raw performance.
- **Make-up exam:** available for officially approved absences (e.g. illness, out-of-school competition).
- **Late-work policy:** not accepted unless officially approved.

4. Lecture Schedule

Week	Topic	Note
1	Introduction to Machine Learning	
2	Regression: Linear & Logistic Regression	
3	Generative Classifiers: Naive Bayes, GDA	
4	Distance-based Models: kNN, k-means, GMM/EM	
5	Support Vector Machines and Kernels	
6	Regularization and Model Selection	
7	Tree-based Models: Decision Trees to GBDT	
8	Block A Capstone: Team Project	Midterm exam
9	Neural Network Basics, Backpropagation	
10	Convolutional Neural Networks	
11	Sequence Models (RNNs)	
12	Attention and Transformers	
13	Graph Representation Learning	
14	Representation Learning: PCA, word2vec	
15	Latent-Variable Generative Models	
16	Block B Capstone: Team Project	Final exam
17	Large Language Models (wrap-up)	

The exact calendar (start date, midterm/final exam period, holidays) will be announced once the official 2027 academic calendar is confirmed.